

# AI Personalized Message Generator

*Internship Reference Document*

|                |                                                                                                |
|----------------|------------------------------------------------------------------------------------------------|
| Company        | Paper Plane                                                                                    |
| Solution       | AI Personalized Message Generator                                                              |
| What it solves | AI creates birthday wishes, anniversary messages, thank-you notes, and corporate gift messages |
| Why useful     | Adds value to customized gifting and saves customer time                                       |
| Duration       | 01 June 2026 - 30 June 2026 (26 Working Days)                                                  |
| Team size      | 3 Students per team                                                                            |
| Roles          | Student 1 - Frontend   Student 2 - Backend   Student 3 - Testing & Deployment                  |

## Introduction Session

This section covers everything the instructor should explain in the Introduction Video (Day 1). Students must watch and understand this before any coding begins.

### 1. About the Company

Paper Plane is a business operating in personalized gifting, online gift ordering, custom gift workflows, product personalization, packaging inventory, corporate bulk gifting, festival hampers, occasion reminders, returns, and AI gift recommendations. The team needs a clear digital system to manage day-to-day records, customer actions, staff ownership, and timely follow-ups.

For this internship project, students will study the real business flow and build a working prototype named AI Personalized Message Generator. The goal is to convert the given solution idea into a usable software reference implementation that supports the company team in practical operations.

### 2. The Problem They Face

Every day, the company receives customer requests, operational updates, and follow-up requirements that need quick action.

Problem: AI creates birthday wishes, anniversary messages, thank-you notes, and corporate gift messages. At present, this type of work can become slow or scattered when it depends only on calls, WhatsApp messages, spreadsheets, or manual follow-up.

This results in:

- Wasted time for the team
- Slow response to important customers or priority cases
- Missed revenue opportunities because follow-ups are delayed
- No clean system to track progress, ownership, and history

### 3. What We Are Building

We are building AI Personalized Message Generator - a smart working prototype that solves this problem end to end.

How the system works:

- Step 1 - Gift Customer/Corporate Buyer enters or updates information through the online gift order page, custom gift form, product personalization manager, design approval tracker, inventory screen, corporate enquiry portal, occasion calendar, return request page, or AI gift assistant
- Step 2 - Backend APIs validate the input and save it with source, status, owner, and timestamp
- Step 3 - The system applies AI message generation workflow that converts occasion, tone, recipient, and relationship into personalized greeting content
- Step 4 - AI or rule-based logic prepares useful outputs such as recommendations, summaries, alerts, or messages
- Step 5 - Gift Store Admin/Designer/Production Team/Dispatch Team/Corporate Sales Team views records, status, priorities, and action history in the dashboard
- Step 6 - Automated reminders, confirmations, or follow-up messages are generated where required
- Step 7 - The final deployed prototype is tested, documented, and presented with report, PPT, GitHub, and demo video

### 4. Technology Stack

The following technologies will be used in this prototype:

| Layer             | Technology                                            | Role in This Project                                                                                                                                                                                                                                                         |
|-------------------|-------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Frontend          | HTML, CSS, JavaScript / React                         | Message generator form, occasion and tone selector, recipient relation field, generated message preview, edit/copy buttons, and saved messages                                                                                                                               |
| Backend           | Node.js Express or Python Flask                       | Prompt APIs, tone templates, message history storage, order/greeting card linkage, and version management endpoints                                                                                                                                                          |
| AI Layer          | OpenAI API / Gemini API or rule-based prompts         | Generate birthday wishes, anniversary notes, thank-you messages, and corporate gifting messages                                                                                                                                                                              |
| Database          | MySQL / PostgreSQL / Firebase                         | Store customers, recipients, gift products, personalization fields, photos, names, messages, design approvals, production stages, packaging materials, corporate enquiries, hampers, important dates, return requests, recommendations, greeting cards, and proposal history |
| Messaging/Reports | WhatsApp Business API / Email API / PDF or CSV export | Send confirmations, reminders, alerts, summaries, and admin reports                                                                                                                                                                                                          |

## 5. Team Structure

Each team has exactly 3 students. The work is divided by role throughout the entire 26-working-day internship:

| Student 1<br>Frontend                                                                                                                          | Student 2<br>Backend                                                                                                | Student 3<br>Testing & Deployment                                                                                                                                                                                                                                                                      |
|------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Message generator form, occasion and tone selector, recipient relation field, generated message preview, edit/copy buttons, and saved messages | Prompt APIs, tone templates, message history storage, order/greeting card linkage, and version management endpoints | Gift ordering testing, customization workflow testing, design approval testing, inventory testing, corporate enquiry testing, hamper builder testing, occasion reminder testing, return workflow testing, AI recommendation/message testing, API testing, GitHub management, and deployment validation |

## 6. What Students Will Build by End of 26 Working Days

- A working frontend for AI Personalized Message Generator connected to backend APIs
- A backend system that stores records, validates inputs, and returns useful status/output
- An AI or rule-based logic module that generates recommendations, summaries, alerts, or workflow outputs
- A personalized gift ordering and fulfilment dashboard showing records, statuses, actions, reports, and counts
- A deployed prototype accessible through a URL
- Full project documentation, PPT presentation, GitHub repository, and demo video

## 7. Review Milestones

There are 3 mandatory review presentations. Missing any review without valid reason = FAIL in internship evaluation.

| Review          | Date            | What to Present                                                           |
|-----------------|-----------------|---------------------------------------------------------------------------|
| <b>Review 1</b> | 06 June 2026    | Company intro, project title, problem statement, objectives, abstract     |
| <b>Review 2</b> | 19-20 June 2026 | Literature survey, system analysis, architecture, DB design, initial code |
| <b>Review 3</b> | 29-30 June 2026 | Full working prototype demo, report, PPT, GitHub, demo video              |



# Day-wise Plan

Each working day is divided into 3 role-specific task columns. Student 1 = Frontend, Student 2 = Backend, Student 3 = Testing & Deployment. All three work in parallel.

## Day 1

### Week 1

01 June 2026 (Mon) - Introduction Session

The instructor explains the company background, project overview, problem statement, technology stack, team roles, 26-working-day plan, and review milestones for AI Personalized Message Generator. No coding today; the goal is complete understanding.

Tasks - All 3 Students

#### Student 1 - Frontend

- Watch and re-read the Introduction Section
- Understand the frontend modules: Message generator form, occasion and tone selector, recipient relation field, generated message preview, edit/copy buttons, and saved messages
- Sketch 3 screens on paper: input screen, workflow/assistant screen, dashboard
- Start Day 1 logbook entry

#### Student 2 - Backend

- Watch and re-read the Introduction Section
- Understand backend responsibilities: Prompt APIs, tone templates, message history storage, order/greeting card linkage, and version management endpoints
- Write 5 backend questions about APIs, database, business rules, and deployment
- Start Day 1 logbook entry

#### Student 3 - Testing & Deployment

- Watch and re-read the Introduction Section
- Research what testing means in a software project
- Research GitHub and why version control is used
- Create GitHub account if needed and start Day 1 logbook entry

#### Deliverable

Groups formed. Logbook Day 1 entries written. All students understand project overview.

## Day 2

### Week 1

02 June 2026 (Tue) - Problem Statement + Abstract

Students document the business problem and write a project abstract for AI Personalized Message Generator. These are Review 1 deliverables.

Tasks - All 3 Students

#### Student 1 - Frontend

- Draft the problem statement in 4-5 sentences from the user interface perspective
- List screens needed for Gift Customer/Corporate Buyer and Gift Store Admin/Designer/Production Team/Dispatch Team/Corporate Sales Team
- Write abstract focusing on user value and business impact

#### Student 2 - Backend

- Draft the problem statement from technical/system perspective
- List backend operations: create, read, update status, process logic, dashboard reports
- Write abstract explaining APIs, database, and logic flow

#### Student 3 - Testing & Deployment

- Create team GitHub repository
- Set up folders: /frontend, /backend, /docs, /tests
- Push README.md with title, company, team names, and tech stack

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| <b>Deliverable</b>                                                                                                                                                                                                                                                                                                                                                  |               | Problem Statement draft. Project Abstract draft. GitHub repository created.          |
| <b>Day 3</b>                                                                                                                                                                                                                                                                                                                                                        | <b>Week 1</b> | <b>03 June 2026 (Wed) - Project Objectives + Tech Stack Setup</b>                    |
| Students write objectives and prepare development environment for AI Personalized Message Generator.                                                                                                                                                                                                                                                                |               |                                                                                      |
| Tasks - All 3 Students                                                                                                                                                                                                                                                                                                                                              |               |                                                                                      |
| <b>Student 1 - Frontend</b>                                                                                                                                                                                                                                                                                                                                         |               |                                                                                      |
| <ul style="list-style-type: none"> <li>- Write 5 objectives from user/frontend perspective</li> <li>- Install VS Code and Node.js</li> <li>- Create frontend folder and first HTML/React screen</li> </ul>                                                                                                                                                          |               |                                                                                      |
| <b>Student 2 - Backend</b>                                                                                                                                                                                                                                                                                                                                          |               |                                                                                      |
| <ul style="list-style-type: none"> <li>- Write 5 objectives from backend/system perspective</li> <li>- Install Node.js Express or Python Flask setup</li> <li>- Create backend server with one test route</li> </ul>                                                                                                                                                |               |                                                                                      |
| <b>Student 3 - Testing &amp; Deployment</b>                                                                                                                                                                                                                                                                                                                         |               |                                                                                      |
| <ul style="list-style-type: none"> <li>- Write 5 objectives from testing and quality perspective</li> <li>- Install Postman for API testing</li> <li>- Test backend Hello route and record result in test log</li> </ul>                                                                                                                                            |               |                                                                                      |
| <b>Deliverable</b>                                                                                                                                                                                                                                                                                                                                                  |               | 5 Project Objectives written. Development environment set up. First test documented. |
| <b>Day 4</b>                                                                                                                                                                                                                                                                                                                                                        | <b>Week 1</b> | <b>04 June 2026 (Thu) - Use Case Design + Wireframes</b>                             |
| Students design use cases and wireframes before coding major screens for AI Personalized Message Generator.                                                                                                                                                                                                                                                         |               |                                                                                      |
| Tasks - All 3 Students                                                                                                                                                                                                                                                                                                                                              |               |                                                                                      |
| <b>Student 1 - Frontend</b>                                                                                                                                                                                                                                                                                                                                         |               |                                                                                      |
| <ul style="list-style-type: none"> <li>- Draw wireframes for input form, processing/workflow screen, dashboard, detail page</li> <li>- Use paper or Figma free version</li> <li>- Upload wireframe images to /docs folder</li> </ul>                                                                                                                                |               |                                                                                      |
| <b>Student 2 - Backend</b>                                                                                                                                                                                                                                                                                                                                          |               |                                                                                      |
| <ul style="list-style-type: none"> <li>- Draw use case diagram with actors: Gift Customer/Corporate Buyer, Gift Store Admin/Designer/Production Team/Dispatch Team/Corporate Sales Team, Admin</li> <li>- Write endpoint list: POST/create, GET/list, GET/detail, POST/process, GET/dashboard</li> <li>- Define request and response fields for each API</li> </ul> |               |                                                                                      |
| <b>Student 3 - Testing &amp; Deployment</b>                                                                                                                                                                                                                                                                                                                         |               |                                                                                      |
| <ul style="list-style-type: none"> <li>- Review wireframes and use case diagram for completeness</li> <li>- Write initial test cases for each use case</li> <li>- Push test case document to /docs</li> </ul>                                                                                                                                                       |               |                                                                                      |
| <b>Deliverable</b>                                                                                                                                                                                                                                                                                                                                                  |               | Wireframes, use case diagram, API endpoint list, and test cases completed.           |

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| <b>Day 5</b>                                                              | <b>Week 1</b> | <b>05 June 2026 (Fri) - Review 1 Preparation</b> |
| All three students prepare Review 1 PPT and complete a mock presentation. |               |                                                  |

Tasks - All 3 Students

#### Student 1 - Frontend

- Build slides: company intro, project title, user screens, wireframes
- Keep slides clean and not overcrowded
- Present frontend sections in team mock

#### Student 2 - Backend

- Build slides: problem statement, proposed system, technology stack
- Prepare simple explanation of backend approach
- Present backend sections in team mock

#### Student 3 - Testing & Deployment

- Build slides: objectives, abstract, GitHub repo link, test plan overview
- Verify all documents are pushed to GitHub
- Present testing sections in team mock

#### Deliverable

Review 1 PPT slides completed and practiced. All docs pushed to GitHub.

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| <b>Day 6</b>                                                                                                                         | <b>Week 1</b> | <b>06 June 2026 (Sat) - REVIEW PRESENTATION 1</b> |
| Team presents company intro, project title, problem statement, objectives, abstract, wireframes, GitHub setup, and technology stack. |               |                                                   |

Tasks - All 3 Students

#### Student 1 - Frontend

- Present wireframes and UI plan
- Answer questions about frontend choices
- Note all feedback received

#### Student 2 - Backend

- Present problem statement, proposed system, and tech stack
- Answer questions about backend approach
- Note all feedback received

#### Student 3 - Testing & Deployment

- Present GitHub repository, objectives, and test case plan
- Answer questions on testing strategy
- Note all feedback received

#### Deliverable

Review 1 completed. Feedback documented. Score received.

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| <b>Day 7</b><br><br>Team applies Review 1 feedback and starts research for AI Personalized Message Generator.                                                                                                                                                                                                         | <b>Week 2</b> | <b>08 June 2026 (Mon) - Review 1 Feedback + Week 2 Prep</b>              |
| Tasks - All 3 Students                                                                                                                                                                                                                                                                                                |               |                                                                          |
| <b>Student 1 - Frontend</b> <ul style="list-style-type: none"> <li>- Update wireframes based on feedback</li> <li>- Research similar UI patterns related to personalized gift ordering and fulfilment dashboard</li> <li>- Begin HTML/React layout for main input screen</li> </ul>                                   |               |                                                                          |
| <b>Student 2 - Backend</b> <ul style="list-style-type: none"> <li>- Update problem statement and objectives based on feedback</li> <li>- Research backend patterns for similar systems</li> <li>- Begin reading REST API structure and validation rules</li> </ul>                                                    |               |                                                                          |
| <b>Student 3 - Testing &amp; Deployment</b> <ul style="list-style-type: none"> <li>- Update test case document based on feedback</li> <li>- Create test tracker: Test ID, Description, Expected, Actual, Pass/Fail</li> <li>- Collect 2-3 references for literature survey</li> </ul>                                 |               |                                                                          |
| <b>Deliverable</b>                                                                                                                                                                                                                                                                                                    |               | Review 1 corrections completed. Week 2 research started.                 |
| <b>Day 8</b><br><br>Students research existing tools and document how AI Personalized Message Generator is different.                                                                                                                                                                                                 | <b>Week 2</b> | <b>09 June 2026 (Tue) - Literature Survey + Existing System Analysis</b> |
| Tasks - All 3 Students                                                                                                                                                                                                                                                                                                |               |                                                                          |
| <b>Student 1 - Frontend</b> <ul style="list-style-type: none"> <li>- Study 2 existing products or websites with similar personalized gift ordering and fulfilment dashboard features</li> <li>- Note dashboard layout, filters, status indicators, and limitations</li> <li>- Write UI/UX comparison notes</li> </ul> |               |                                                                          |
| <b>Student 2 - Backend</b> <ul style="list-style-type: none"> <li>- Study 2-3 existing systems from backend/workflow perspective</li> <li>- Write existing system analysis with features and limitations</li> <li>- Define the gap filled by the proposed system</li> </ul>                                           |               |                                                                          |
| <b>Student 3 - Testing &amp; Deployment</b> <ul style="list-style-type: none"> <li>- Summarise 3 references in 5 bullet points each</li> <li>- Add references to literature survey document</li> <li>- Check source names and links are clearly mentioned</li> </ul>                                                  |               |                                                                          |
| <b>Deliverable</b>                                                                                                                                                                                                                                                                                                    |               | Existing System Analysis document. Literature Survey with 3+ references. |

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| <b>Day 9</b><br><br>Students describe the proposed system and design database tables for AI Personalized Message Generator.                                                                                                                                                                                                                                                                                                                                                      | <b>Week 2</b>                                                        | <b>10 June 2026 (Wed) - Proposed System Description + DB Design</b> |
| Tasks - All 3 Students                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                      |                                                                     |
| <b>Student 1 - Frontend</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                      |                                                                     |
| <ul style="list-style-type: none"> <li>- Write proposed system description for each screen</li> <li>- Create comparison table: Existing System vs Proposed System</li> <li>- Refine wireframes based on final proposed flow</li> </ul>                                                                                                                                                                                                                                           |                                                                      |                                                                     |
| <b>Student 2 - Backend</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                      |                                                                     |
| <ul style="list-style-type: none"> <li>- Design database schema to store customers, recipients, gift products, personalization fields, photos, names, messages, design approvals, production stages, packaging materials, corporate enquiries, hampers, important dates, return requests, recommendations, greeting cards, and proposal history</li> <li>- Create ER diagram with relationships and key fields</li> <li>- Document validation rules and status values</li> </ul> |                                                                      |                                                                     |
| <b>Student 3 - Testing &amp; Deployment</b>                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                      |                                                                     |
| <ul style="list-style-type: none"> <li>- Write test plan mapped to each feature</li> <li>- Review ER diagram for missing fields</li> <li>- Push ER diagram and test plan to GitHub</li> </ul>                                                                                                                                                                                                                                                                                    |                                                                      |                                                                     |
| <b>Deliverable</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Proposed System Description. ER Diagram. Database schema. Test plan. |                                                                     |

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| <b>Day 10</b><br><br>Backend database is created and the first API route works. Frontend builds the main form for AI Personalized Message Generator.                                                   | <b>Week 2</b>                                                     | <b>11 June 2026 (Thu) - Database Setup + First Backend Route + Main Form</b> |
| Tasks - All 3 Students                                                                                                                                                                                 |                                                                   |                                                                              |
| <b>Student 1 - Frontend</b>                                                                                                                                                                            |                                                                   |                                                                              |
| <ul style="list-style-type: none"> <li>- Build full form structure with required fields</li> <li>- Apply basic CSS matching wireframe</li> <li>- Test form layout in browser</li> </ul>                |                                                                   |                                                                              |
| <b>Student 2 - Backend</b>                                                                                                                                                                             |                                                                   |                                                                              |
| <ul style="list-style-type: none"> <li>- Set up MySQL/PostgreSQL/Firebase database</li> <li>- Create main table using Day 9 schema</li> <li>- Build POST/create route and test with Postman</li> </ul> |                                                                   |                                                                              |
| <b>Student 3 - Testing &amp; Deployment</b>                                                                                                                                                            |                                                                   |                                                                              |
| <ul style="list-style-type: none"> <li>- Test POST/create with valid data</li> <li>- Test missing fields and invalid data</li> <li>- Document results in test tracker</li> </ul>                       |                                                                   |                                                                              |
| <b>Deliverable</b>                                                                                                                                                                                     | Database table created. POST/create API working. Main form built. |                                                                              |



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| <b>Day 11</b>                                                                                                | <b>Week 2</b> | <b>12 June 2026 (Fri) - List API + Dashboard HTML</b> |
| Backend adds list route. Frontend creates static personalized gift ordering and fulfilment dashboard layout. |               |                                                       |

Tasks - All 3 Students

#### Student 1 - Frontend

- Build dashboard HTML/React structure with cards/table view
- Add columns: name/title, status, date, owner/amount/score as applicable
- Use dummy records to verify layout

#### Student 2 - Backend

- Build GET/list route returning all records as JSON
- Add simple filters such as status/type/date
- Test list and filter routes in Postman

#### Student 3 - Testing & Deployment

- Test GET/list JSON structure
- Test filters with sample data
- Push tested code to GitHub

#### Deliverable

GET/list API working. Dashboard HTML structure complete. Tests passing.

|                                                                            |               |                                                                      |
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| <b>Day 12</b>                                                              | <b>Week 2</b> | <b>13 June 2026 (Sat) - GitHub Workflow + Environment Validation</b> |
| Team consolidates code into shared GitHub repo and validates merged setup. |               |                                                                      |

Tasks - All 3 Students

#### Student 1 - Frontend

- Create feature/frontend branch
- Push frontend form and dashboard files
- Raise pull request to main

#### Student 2 - Backend

- Create feature/backend-api branch
- Push backend routes and database scripts
- Raise pull request to main

#### Student 3 - Testing & Deployment

- Review pull requests for naming, comments, and structure
- Test merged main branch end-to-end locally
- Update test tracker with merge validation

#### Deliverable

GitHub branches created. PRs raised and merged. Main branch stable and tested.

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| <b>Day 13</b><br><br>Team designs the core logic before deeper implementation for AI Personalized Message Generator.                                                                                                                                                                                                                       | <b>Week 3</b>                                                                     | <b>15 June 2026 (Mon) - Core Logic Design + Workflow Plan</b>            |
| Tasks - All 3 Students                                                                                                                                                                                                                                                                                                                     |                                                                                   |                                                                          |
| <b>Student 1 - Frontend</b> <ul style="list-style-type: none"> <li>- Design workflow screen or assistant flow based on user journey</li> <li>- Write exact labels/messages shown to users</li> <li>- Create static mockup for the workflow screen</li> </ul>                                                                               |                                                                                   |                                                                          |
| <b>Student 2 - Backend</b> <ul style="list-style-type: none"> <li>- Define AI message generation workflow that converts occasion, tone, recipient, and relationship into personalized greeting content in a clear rule document</li> <li>- Write processing function or service module</li> <li>- Test logic with sample inputs</li> </ul> |                                                                                   |                                                                          |
| <b>Student 3 - Testing &amp; Deployment</b> <ul style="list-style-type: none"> <li>- Prepare 10 sample test cases for core logic</li> <li>- Review edge cases and skipped/invalid inputs</li> <li>- Document expected output for each case</li> </ul>                                                                                      |                                                                                   |                                                                          |
| <b>Deliverable</b>                                                                                                                                                                                                                                                                                                                         | Core logic document. Working processing function. Workflow UI mockup. Test cases. |                                                                          |
| <b>Day 14</b><br><br>Week 2 documents and Review 2 presentation are finalised.                                                                                                                                                                                                                                                             | <b>Week 3</b>                                                                     | <b>16 June 2026 (Tue) - Week 2 Review Prep + Literature Survey Final</b> |
| Tasks - All 3 Students                                                                                                                                                                                                                                                                                                                     |                                                                                   |                                                                          |
| <b>Student 1 - Frontend</b> <ul style="list-style-type: none"> <li>- Finalise frontend files: form, dashboard, workflow mockup</li> <li>- Prepare slides: wireframes, UI design, comparison table</li> <li>- Do mock presentation</li> </ul>                                                                                               |                                                                                   |                                                                          |
| <b>Student 2 - Backend</b> <ul style="list-style-type: none"> <li>- Finalise backend files: schema, APIs, processing logic</li> <li>- Prepare slides: architecture, DB design, core logic</li> <li>- Do mock presentation</li> </ul>                                                                                                       |                                                                                   |                                                                          |
| <b>Student 3 - Testing &amp; Deployment</b> <ul style="list-style-type: none"> <li>- Finalise literature survey with 5+ references</li> <li>- Prepare slides: test plan, existing system analysis, GitHub status</li> <li>- Ensure repo is clean and updated</li> </ul>                                                                    |                                                                                   |                                                                          |
| <b>Deliverable</b>                                                                                                                                                                                                                                                                                                                         | Literature survey final. Week 2 documents complete. Review 2 PPT ready.           |                                                                          |

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| <b>Day 15</b><br><br>Students create architecture diagram and integration plan for AI Personalized Message Generator.                                                                                                                                               | <b>Week 3</b> | <b>17 June 2026 (Wed) - System Architecture Diagram + Full Stack Plan</b> |
| Tasks - All 3 Students                                                                                                                                                                                                                                              |               |                                                                           |
| <b>Student 1 - Frontend</b> <ul style="list-style-type: none"> <li>- Create architecture diagram using draw.io</li> <li>- Label frontend, backend API, database, AI/rule engine, notification/reporting service</li> <li>- Export image and add to /docs</li> </ul> |               |                                                                           |
| <b>Student 2 - Backend</b> <ul style="list-style-type: none"> <li>- Review backend components in architecture</li> <li>- Plan how frontend calls backend APIs</li> <li>- Confirm API contracts and response structures</li> </ul>                                   |               |                                                                           |
| <b>Student 3 - Testing &amp; Deployment</b> <ul style="list-style-type: none"> <li>- Review architecture for testability</li> <li>- Create integration test plan</li> <li>- Push architecture and test plan to GitHub</li> </ul>                                    |               |                                                                           |
| <b>Deliverable</b>                                                                                                                                                                                                                                                  |               | Final System Architecture Diagram. Integration test plan.                 |
| <b>Day 16</b><br><br>Main form now sends real data to backend. This is the first end-to-end integration.                                                                                                                                                            | <b>Week 3</b> | <b>18 June 2026 (Thu) - Connect Frontend Form to Backend API</b>          |
| Tasks - All 3 Students                                                                                                                                                                                                                                              |               |                                                                           |
| <b>Student 1 - Frontend</b> <ul style="list-style-type: none"> <li>- Add fetch/axios call from form to POST API</li> <li>- Show success message after submit</li> <li>- Show client-side validation errors</li> </ul>                                               |               |                                                                           |
| <b>Student 2 - Backend</b> <ul style="list-style-type: none"> <li>- Return clean JSON response from POST API</li> <li>- Add server-side validation for required fields</li> <li>- Test from browser and Postman</li> </ul>                                          |               |                                                                           |
| <b>Student 3 - Testing &amp; Deployment</b> <ul style="list-style-type: none"> <li>- Test full form submit flow</li> <li>- Check database record creation</li> <li>- Test empty fields, long text, and special characters</li> </ul>                                |               |                                                                           |
| <b>Deliverable</b>                                                                                                                                                                                                                                                  |               | Frontend form connected to backend API. End-to-end create flow working.   |

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| Day 17                                                                                                                                                                                                                   | Week 3 | 19 June 2026 (Fri) - REVIEW<br>PRESENTATION 2 - Day 1                      |
| First batch presents literature survey, existing system analysis, architecture, database design, and initial working code.                                                                                               |        |                                                                            |
| Tasks - All 3 Students                                                                                                                                                                                                   |        |                                                                            |
| Student 1 - Frontend                                                                                                                                                                                                     |        |                                                                            |
| <ul style="list-style-type: none"> <li>- Present wireframes, workflow UI, dashboard screenshots</li> <li>- Show live form in browser</li> <li>- Answer questions on UI decisions</li> </ul>                              |        |                                                                            |
| Student 2 - Backend                                                                                                                                                                                                      |        |                                                                            |
| <ul style="list-style-type: none"> <li>- Present architecture diagram, database schema, and core logic</li> <li>- Show backend routes working in Postman</li> <li>- Answer technical implementation questions</li> </ul> |        |                                                                            |
| Student 3 - Testing & Deployment                                                                                                                                                                                         |        |                                                                            |
| <ul style="list-style-type: none"> <li>- Present test plan, existing system analysis, GitHub repository</li> <li>- Show current test results</li> <li>- Answer testing and quality questions</li> </ul>                  |        |                                                                            |
| Deliverable                                                                                                                                                                                                              |        | Review 2 Day 1 completed. Scores received. Feedback documented.            |
| Day 18                                                                                                                                                                                                                   | Week 3 | 20 June 2026 (Sat) - REVIEW<br>PRESENTATION 2 - Day 2 + Core Feature Build |
| Remaining teams present Review 2. After presentations, teams continue building the main feature flow.                                                                                                                    |        |                                                                            |
| Tasks - All 3 Students                                                                                                                                                                                                   |        |                                                                            |
| Student 1 - Frontend                                                                                                                                                                                                     |        |                                                                            |
| <ul style="list-style-type: none"> <li>- Present Review 2 if in second batch</li> <li>- Build status/action section in dashboard</li> <li>- Add UI for history or generated output where needed</li> </ul>               |        |                                                                            |
| Student 2 - Backend                                                                                                                                                                                                      |        |                                                                            |
| <ul style="list-style-type: none"> <li>- Present Review 2 if in second batch</li> <li>- Build POST/process or status-update route</li> <li>- Save processing result and history in database</li> </ul>                   |        |                                                                            |
| Student 3 - Testing & Deployment                                                                                                                                                                                         |        |                                                                            |
| <ul style="list-style-type: none"> <li>- Present Review 2 if in second batch</li> <li>- Test process/status route</li> <li>- Write 5 additional test cases for main logic</li> </ul>                                     |        |                                                                            |
| Deliverable                                                                                                                                                                                                              |        | Review 2 Day 2 completed. Main processing/status route started.            |

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| <b>Day 19</b><br><br>The main AI Personalized Message Generator workflow is completed with alerts, output/history, or status updates.                                                                                    | <b>Week 4</b> | <b>22 June 2026 (Mon) - Core Workflow Complete + Alerts/Outputs</b>             |
| Tasks - All 3 Students                                                                                                                                                                                                   |               |                                                                                 |
| <b>Student 1 - Frontend</b>                                                                                                                                                                                              |               |                                                                                 |
| <ul style="list-style-type: none"> <li>- Show generated output/status/result on dashboard</li> <li>- Add badges or alerts for priority records</li> <li>- Add refresh button or auto-refresh</li> </ul>                  |               |                                                                                 |
| <b>Student 2 - Backend</b>                                                                                                                                                                                               |               |                                                                                 |
| <ul style="list-style-type: none"> <li>- Complete processing trigger and status update logic</li> <li>- Build notification/reporting endpoint if required</li> <li>- Store output, result, and action history</li> </ul> |               |                                                                                 |
| <b>Student 3 - Testing &amp; Deployment</b>                                                                                                                                                                              |               |                                                                                 |
| <ul style="list-style-type: none"> <li>- Test full pipeline from input to output/dashboard</li> <li>- Test different record types and statuses</li> <li>- Document all results and bugs</li> </ul>                       |               |                                                                                 |
| <b>Deliverable</b>                                                                                                                                                                                                       |               | Core workflow fully working. Alerts/outputs visible. Pipeline tests documented. |

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| <b>Day 20</b><br><br>A detail page is built and full integration testing runs across the entire system.                                                                                                                                     | <b>Week 4</b> | <b>23 June 2026 (Tue) - Detail Page + Full Integration Test</b>            |
| Tasks - All 3 Students                                                                                                                                                                                                                      |               |                                                                            |
| <b>Student 1 - Frontend</b>                                                                                                                                                                                                                 |               |                                                                            |
| <ul style="list-style-type: none"> <li>- Build detail page opened from dashboard item</li> <li>- Show complete record details, status, notes, output, and history</li> <li>- Add Back button to return to dashboard</li> </ul>              |               |                                                                            |
| <b>Student 2 - Backend</b>                                                                                                                                                                                                                  |               |                                                                            |
| <ul style="list-style-type: none"> <li>- Build GET/detail route</li> <li>- Return main record with related history/output data</li> <li>- Verify joins or linked records are correct</li> </ul>                                             |               |                                                                            |
| <b>Student 3 - Testing &amp; Deployment</b>                                                                                                                                                                                                 |               |                                                                            |
| <ul style="list-style-type: none"> <li>- Run full integration test: create -&gt; process -&gt; dashboard -&gt; detail</li> <li>- Test at least 5 records with different statuses</li> <li>- Document all results and report bugs</li> </ul> |               |                                                                            |
| <b>Deliverable</b>                                                                                                                                                                                                                          |               | Detail Page working. GET/detail API working. Full integration test passed. |

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| <b>Day 21</b><br><br>The prototype gets a polish pass and clear error handling is added.                                                                                                                                        | <b>Week 4</b> | <b>24 June 2026 (Wed) - UI Polish + Error Handling</b>                                           |
| Tasks - All 3 Students                                                                                                                                                                                                          |               |                                                                                                  |
| <b>Student 1 - Frontend</b>                                                                                                                                                                                                     |               |                                                                                                  |
| <ul style="list-style-type: none"> <li>- Clean CSS: consistent fonts, spacing, colors, and responsive layout</li> <li>- Add loading state for submit/dashboard load</li> <li>- Add empty state when no records exist</li> </ul> |               |                                                                                                  |
| <b>Student 2 - Backend</b>                                                                                                                                                                                                      |               |                                                                                                  |
| <ul style="list-style-type: none"> <li>- Add try/catch error handling to all backend routes</li> <li>- Return meaningful errors: 400 for bad input, 500 for server error</li> <li>- Add console logs for debugging</li> </ul>   |               |                                                                                                  |
| <b>Student 3 - Testing &amp; Deployment</b>                                                                                                                                                                                     |               |                                                                                                  |
| <ul style="list-style-type: none"> <li>- Test bad input, missing data, failed API, and empty dashboard</li> <li>- Verify frontend error messages are clear</li> <li>- Update tracker with error handling tests</li> </ul>       |               |                                                                                                  |
| <b>Deliverable</b>                                                                                                                                                                                                              |               | Polished UI. Error handling on all routes. Error test cases passing.                             |
| <b>Day 22</b><br><br>The prototype is deployed so it can be accessed through a URL instead of localhost.                                                                                                                        | <b>Week 4</b> | <b>25 June 2026 (Thu) - Deployment Setup</b>                                                     |
| Tasks - All 3 Students                                                                                                                                                                                                          |               |                                                                                                  |
| <b>Student 1 - Frontend</b>                                                                                                                                                                                                     |               |                                                                                                  |
| <ul style="list-style-type: none"> <li>- Prepare frontend for deployment</li> <li>- Update API URLs to deployed backend URL</li> <li>- Take screenshots of all screens for report</li> </ul>                                    |               |                                                                                                  |
| <b>Student 2 - Backend</b>                                                                                                                                                                                                      |               |                                                                                                  |
| <ul style="list-style-type: none"> <li>- Deploy backend to Render/Railway or equivalent free tier</li> <li>- Deploy database or configure Firebase</li> <li>- Test all API routes at deployed URL</li> </ul>                    |               |                                                                                                  |
| <b>Student 3 - Testing &amp; Deployment</b>                                                                                                                                                                                     |               |                                                                                                  |
| <ul style="list-style-type: none"> <li>- Test deployed version end-to-end</li> <li>- Document deployment URL and test results</li> <li>- Create deployment guide document</li> </ul>                                            |               |                                                                                                  |
| <b>Deliverable</b>                                                                                                                                                                                                              |               | Backend deployed. Database deployed. Frontend connected to production. Deployment guide written. |

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| <b>Day 23</b><br><br>Deployed AI Personalized Message Generator is tested end-to-end. Bug fixes are completed and report writing begins.                                                                                                                       | <b>Week 4</b> | <b>26 June 2026 (Fri) - Final Testing + Bug Fixes + Report Part 1</b>                    |
| Tasks - All 3 Students                                                                                                                                                                                                                                         |               |                                                                                          |
| <b>Student 1 - Frontend</b> <ul style="list-style-type: none"> <li>- Test all screens on deployed version</li> <li>- Fix responsive layout issues</li> <li>- Write Introduction chapter and UI design section</li> </ul>                                       |               |                                                                                          |
| <b>Student 2 - Backend</b> <ul style="list-style-type: none"> <li>- Review deployed backend routes</li> <li>- Verify database queries with real data</li> <li>- Write System Analysis and Implementation chapters</li> </ul>                                   |               |                                                                                          |
| <b>Student 3 - Testing &amp; Deployment</b> <ul style="list-style-type: none"> <li>- Run final test suite on deployed URL</li> <li>- Create bug report with fixed and known issues</li> <li>- Write Literature Survey and Testing chapter draft</li> </ul>     |               |                                                                                          |
| <b>Deliverable</b>                                                                                                                                                                                                                                             |               | Final test suite results. Bug report. Report Part 1 draft completed.                     |
| <b>Day 24</b><br><br>Report, demo video, final PPT, and mock presentation are completed before final review.                                                                                                                                                   | <b>Week 4</b> | <b>27 June 2026 (Sat) - Project Report Part 2 + Demo Video + Final PPT</b>               |
| Tasks - All 3 Students                                                                                                                                                                                                                                         |               |                                                                                          |
| <b>Student 1 - Frontend</b> <ul style="list-style-type: none"> <li>- Record 3-5 minute frontend demo</li> <li>- Upload demo video to Google Drive/YouTube unlisted</li> <li>- Build PPT slides: Title, Introduction, Problem, Wireframes, Live Demo</li> </ul> |               |                                                                                          |
| <b>Student 2 - Backend</b> <ul style="list-style-type: none"> <li>- Write Conclusion and References</li> <li>- Review report for technical accuracy</li> <li>- Build PPT slides: Architecture, DB design, APIs, logic</li> </ul>                               |               |                                                                                          |
| <b>Student 3 - Testing &amp; Deployment</b> <ul style="list-style-type: none"> <li>- Finalise Testing chapter</li> <li>- Proof-read report and export PDF</li> <li>- Build PPT slides: Testing, Bug report, Deployment, Conclusion, Future Scope</li> </ul>    |               |                                                                                          |
| <b>Deliverable</b>                                                                                                                                                                                                                                             |               | Complete Project Report PDF. Demo video linked. Final PPT ready. Mock presentation done. |

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| <b>Day 25</b>                                                                                                                                                                                                      | <b>Week 5</b>                                                          | <b>29 June 2026 (Mon) - REVIEW<br/>PRESENTATION 3 - Day 1 + Final<br/>Submissions</b> |
| First batch delivers final review with live demo and submits all deliverables.                                                                                                                                     |                                                                        |                                                                                       |
| Tasks - All 3 Students                                                                                                                                                                                             |                                                                        |                                                                                       |
| <b>Student 1 - Frontend</b>                                                                                                                                                                                        |                                                                        |                                                                                       |
| <ul style="list-style-type: none"> <li>- Present frontend live demo and screens</li> <li>- Answer questions on UI decisions and implementation</li> </ul>                                                          |                                                                        |                                                                                       |
| <b>Student 2 - Backend</b>                                                                                                                                                                                         |                                                                        |                                                                                       |
| <ul style="list-style-type: none"> <li>- Present backend architecture, APIs, database, and logic</li> <li>- Answer technical implementation questions</li> </ul>                                                   |                                                                        |                                                                                       |
| <b>Student 3 - Testing &amp; Deployment</b>                                                                                                                                                                        |                                                                        |                                                                                       |
| <ul style="list-style-type: none"> <li>- Present testing results, deployment URL, GitHub, report PDF, and demo video</li> <li>- Submit final checklist: project, GitHub, PPT, report, demo video, diary</li> </ul> |                                                                        |                                                                                       |
| <b>Deliverable</b>                                                                                                                                                                                                 | Review 3 Day 1 complete. Final deliverables submitted for first batch. |                                                                                       |

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| <b>Day 26</b>                                                                                                                                                                                          | <b>Week 5</b>                                                                                   | <b>30 June 2026 (Tue) - REVIEW<br/>PRESENTATION 3 - Day 2 + Internship<br/>Closing</b> |
| Remaining teams deliver final review. Internship closes with team learning discussion.                                                                                                                 |                                                                                                 |                                                                                        |
| Tasks - All 3 Students                                                                                                                                                                                 |                                                                                                 |                                                                                        |
| <b>Student 1 - Frontend</b>                                                                                                                                                                            |                                                                                                 |                                                                                        |
| <ul style="list-style-type: none"> <li>- Present Review 3 if second batch</li> <li>- Share frontend learnings and challenges</li> <li>- Write final logbook entry</li> </ul>                           |                                                                                                 |                                                                                        |
| <b>Student 2 - Backend</b>                                                                                                                                                                             |                                                                                                 |                                                                                        |
| <ul style="list-style-type: none"> <li>- Present Review 3 if second batch</li> <li>- Share backend learnings and technical decisions</li> <li>- Write final logbook entry</li> </ul>                   |                                                                                                 |                                                                                        |
| <b>Student 3 - Testing &amp; Deployment</b>                                                                                                                                                            |                                                                                                 |                                                                                        |
| <ul style="list-style-type: none"> <li>- Present Review 3 if second batch</li> <li>- Share testing/deployment learnings</li> <li>- Write final logbook entry and confirm internship closure</li> </ul> |                                                                                                 |                                                                                        |
| <b>Deliverable</b>                                                                                                                                                                                     | Review 3 Day 2 complete. All deliverables submitted. Project diary complete. Internship closed. |                                                                                        |